EFFECTIVENESS OF ICT TO TRANSFORM TEACHING EXPERIENCE - A STUDY AMONG SCHOOL TEACHERS IN KERALA

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Abstract - "Education is indispensable for a growing society. It has to continuously evolve to fulfil the needs of fast-changing world" - Peter Serdyukov (2017). ICT is often recognised as a "major tool for building knowledge societies", with its multidimensional approach to provide a pathway to reimagine and redesign the educational systems. The power to ICT to act as a catalyst for improving the learning as well as teaching experience is at the core of its competency. This study aims to analyse the effectiveness of ICT to transform teaching experience among the primary school teachers in Kozhikode district in Kerala. Questionnaires were distributed among 132 primary school teachers and the collected data was analysed for descriptive results and SPSS software was used for inferential results. The results of the study clearly indicate that ICT enabled teaching pedagogies have more impact over traditional face-to-face lecture method as itwidens the gateways for collaborative learning. The participants have also identified that technology-enabled learning can solve many of the existing challenges to the educational system and has the potential to save the time and efforts for both teachers and students.

Keywords - ICT integration; Technology enabled learning Teaching and learning; Primary schools; Kerala

I. INTRODUCTION

Technology has undoubtedly transformed the life of people in the 21st century and the education sector is considerably impacted with its onset, considering the relatively traditional methodologies that it had always embraced. Few developed countries began the race a little earlier and now developing and underdeveloped countries are also trying to catch up with this ever-evolving race to embrace and evolve with technology. Integration of technology has gone far to an extent where it totally altered the way people think, work and live (Grabe & Grabe , 2007).

Teaching pedagogies in schools were the battlefields of early implementation of ICT as it became increasingly important to make students, media literate to face the expectations of their college requirements and early job roles. Thus, schools and other primary educational institutions were required to adopt ICT integration into their curriculum to prepare students and thereby build "a knowledge society".

Implementation of ICT require insightful planning and implementation. The goals and objectives to be achieved is the primary thread around which sustainable plans will be drafted. Teachers and policy makers will have to work hand in hand to reach on a common consensus on what needs to be achieved. (Dudeney, 2010) identifies the primary importance of ICT policies in a country is to provide a set of achievable goals and vision on how the educational system in a country should function. What is to be achieved often takes the centre stage as the methods and pathways are framed keeping in mind what is to be achieved. We can thus summarise that, there are multiple factors at play in order to enable ICT based learning in classrooms. From policy changes to upskilling of teachers and proper maintenance of ICT equipment's is integral to this process.

II. ATTITUDES TOWARDS TECHNOLOGY-BASED TEACHING AND LEARNING

Increase in the use of information technology also lead to the increase in dissatisfaction and resistance to adopt technology in public schools around the world (Hu, Clark, & Ma, 2003). Adoption of sufficient technology will enable academicians to become more effective in sharing knowledge with their students. After understanding the reception of public-school teachers in China towards technology, (Hu, Clark, & Ma, 2003)found that teachers mostly used Microsoft PowerPoint for their daily classes.

The expectations from ICT enabled methodology from teachers and students have increased over the years with the high degree of integration of technology into their daily life. The millennials are associated with different types of gadgets right from their early childhood, which has paved way for their increased expectations from classroom learning. A study by (Chien, Wu, & Hsu, 2014) has shown that younger the students, more are their expectations. Usage of different technologies from mobile phones to interactive televisions are also a likely reason for higher expectations during classroom interaction using ICT.

III. INTEGRATION OF ICT IN THE INDIAN CONTEXT

The "Department of School Education and Literacy" which is part of the "Ministry of Human Resource Development" (MHRD) of the Government of India worked in collaboration with the Global E-schools and Communities Initiative (GeSCI) to formulate a draft policy for usage of ICT in school education in India. The objective revolved around the implementation of ICT tools to enhance learning – teaching experience and to address its challenges by

utilizing newer technological tools. A draft policy was prepared and approved by the Central Advisory Board of Education. In December of 2004, the National Policy on ICT Enabled School Education was launched. The policy aimed primarily at preparing the youth of India to be competent globally in alignment with the technological interventions in the learning domain. This plan was further revised on 9th January, 2010 to strengthen the downfalls of the primary model. The new plan insisted on the need for "recognition of quality and equity, setting up of smart schools at district level, a trained ICT coordinator at each school, the need for developing E-content and the need for collaboration with the community that work closely with the schools like school committees, Teachers-Parents association and other local bodies"(Kundu & Nath, 2018). ICT in schools is included as a subset of RashtriyaMadhyamikShikshaAbhiyan (RMSA), which aims at bridging the gap among students from various social, cultural and economic backgrounds in the country. The RMSA scheme also assist in building computer labs and setting up of ICT enabling tools across different states and Union Territories.

Objectives

1. To evaluate the effectiveness of ICT to transform the teaching experience among primary school teachers in the Kozhikode district of Kerala
2. To understand the attitude and perceptionof teachers towards ICT enabled teaching pedagogies
3. To find out the factors motivating teachers to use Technology-Enabled Tools for Teaching
4. To explore the barriers faced by teachers toadopt Technology-Enabled Tools

IV. THEORETICAL FRAMEWORK

As this researchaims at understanding the effectiveness of ICT to transform the teaching experience of school teachers, the conceptual framework adopted is based on Diffusion of Innovations (Rogers, 2003)and Technology Acceptance Model (Davis, 2003). Being one of the oldest social science theories, Diffusion of Innovation tries to understand how abehaviour, idea or productis adopted by a social system to an extend where it becomes part of the same. Adoption here would mean that there is a change in the way people previously thought or consumed, which lead them to adopt the new. The change in attitude is not uniform for all people in a social system, whereby the rate of adoption is different for different people. Further research has also revealed that the characteristics of people who adopt an innovation in the early stages are different from those who adopt it at a later stage. Based on this characteristics, adopters of an innovation can be categorised into five types; Innovators, early adopters, early majority, late majority and laggards. Different strategies are adopted to invoke reception onto these categories as their nature and characteristics are inherently dissimilar.

Technology Acceptance Model (Davis, 2003)tries to "predict the acceptability of an information system" by analysing the intention behind adopting a new innovation. The primary usefulness of this model is to access the acceptability or success of an innovation or technology and to understand the necessary changes to be made in order to make it more acceptable for the prospective adaptors.According to this model acceptability is determined by "perceived usefulness and perceived ease of use".

The above frameworks, Technology Acceptance Model(Davis, 2003)and Diffusion of Innovation(Rogers, 2003)will help the researcher to understand how thescale and speed of adoption will influence the perceived use and ease of adopting ICT. However, the main variable under study will be the acceptance level of ICT by primary school teachers and its perceived effectiveness in teaching.

V. METHODOLOGY

The researcher adopted quantitative methodology to collect and analyse data from the selected respondents. The questionnaire was designed to address the specific research questions and was created keeping in mind the target audiencei.e.,primary school teachers. The questionnaire also intended to understand the effectiveness of ICT to transform the teaching experience from the teacher's perspective. An attitude evaluation towards ICT enabled teaching pedagogies was also an instrumental part of the questionnaire.

Population and Sampling

The sample consisted of 132 primary school teachers from the Kozhikode district of Kerala. The questionnaire was randomly distributed among teachers of Government (N=60), Private-aided (N=40) and Private-non-aided (N=32) schools in the district. The questionnaire was distributed to the chosen sample by disregarding their gender, age, race and years of experience.

Instrument

The questionnaire consisted of 66 items that were used to collect and analyse the data in order to understand the effectiveness of ICT in transforming the teaching experience among primary school teachers in the Kozhikode district of Kerala. A total of 150 questionnaires were distributed online, out of which 132 was only returned. The respondents, i.e. teachers of primary schools across the district of Kozhikode was asked to read and answer the questions on the questionnaire that was divided into

six sections. The section A was aimed at understanding the demographic background of the respondents which included the name, gender, type of school, age, highest qualification, teaching experience and subject handled. The second section included questions to understand the accessibility and familiarity with using ICT tools. Sections 3 and 4, explored the usage and understanding of ICT tools for teaching and learning. Section 5, looked at the training and development given to individual teachers and section 6 – explored the perceptions around the application of ICT as an enabler for enhanced teaching and learning experience. The questionnaire designed by (Das & Mishra, 2016), was altered to fit the research objectives.

Data Collection Procedure

The questionnaire was created on google forms platform and shared to the selected respondents through email and WhatsApp. Having distributed the questionnaire among 150teachers, only 132 responded and have filled their inputs. The data collection spanned across a period of three weeks from the first to the third week of October 2020. Once it was verified that the respondents have filled the questionnaire, the collected data was analysed by the researcher to infer meaning out of it.

Data Analysis Process

The collected data was analysed using SPSS (Statistical Package for the Social Sciences) software. The researcher made descriptive and inferential analysis of the data collected through the questionnaire. The demographic details collected, was analysed using descriptive analysis. The mean, standard deviation, percentage and frequency was analysed to understand the effect of ICT to transform the teaching experience of primary school teachers in Kozhikode district of Kerala.

Results

This section is aimed at making inferences from the data collected through the questionnaire that was shared with the respondents of the study. The sections below follow the same order as that in the questionnaire. Inferential analysis followed by reliability testing is also conducted on the collected data.

FACTORS	FREQUENCY	PERCENTAGE (%)
Gender		
Male	60	45.45
Female	72	54.55
Teaching Experience		
< 4 Years	15	11.36
5 Years to 15 Years	37	28.03
16 Years to 25 Years	46	34.85
26 Years to 35 Years	32	24.24
> 35	2	1.52
Type of School		
Government School	52	39.39
Aided	64	48.48
Private and Unaided	16	12.12
Current Teaching Position		
Permanent Teacher	114	86.36
Principal / Headmaster	11	8.33
Temporary / Special Teacher	7	5.30
Nature of the classes taken		
Traditionalface-to-face (Lecturing method)	48	36.36
Completely online	11	8.33
Blended method	73	55.3

Table 1. Demographic details of respondents

The total population consisted of 132 respondents who were teachers of primary schools across the Kozhikode district in Kerala. The overall population included 60 male respondents which consumed 45.45% of the sample and the rest 72 (54.55%) were female respondents. Segregating based on teaching experience, the populated consisted of 46 (34.85%) teachers who had 16 to 25 years of experience, 37 (28.03%) respondents were with 5 to 15 years, 32 (24.24%) with 26 to 35 years of experience, 15 (11.36%) with less than 4 years and 2 (1.52%) with more than 35 years of teaching experience. The total population included 114 (86.36%) permanent teachers, 11 (8.33%) of them were either a Headmaster or Principal of a school and 7 (5.30%) of the respondents were special or temporary teachers. The survey also threw light on the regular nature of classes taken by the population which shows that 73

(55.3%) of the respondents relied on blended methodologies for instruction in their classes, 48 (36.36%) used lecturing methods and 11 (8.33%) used full online method of teaching.

Access to Digital Tools

FACTORS	FREQUENCY	PERCENTAGE (%)
Do you own any of these devices?		
Desktop computer	54	40.91
Laptop	89	67.42
Smartphone	127	96.21
Tablet device	24	18.18
Internet is accessed through?		
Mobile devices	103	78.03
Broadband Internet	57	43.18
Wireless Devices	16	12.12
Device used to most frequently access the Internet?		
Smartphone	105	79.55
Desktop computer	3	2.27
Laptop or Tablet	24	18.18
Frequency of Internet usage		
Daily	119	90.15
Once a week	7	5.3
Rarely	6	4.55

Table 2. Accessibility of teachers to digital tools

The table 2, includes the questions that are aimed to understand the accessibility of the teachers towards different digital tools. The survey reveals that, 127 (96.21%) of the total population owns a smartphone, 89 (67.42%) owns a laptop, 54 (40.91%) of the total respondents own a desktop computer and only 24 (18.18%) own a tablet. Out of the total population of 132 teachers, 103 (78.03%) of them have access to internet through their mobile devices, whereas 57 (43.18%) have a broadband connection at their home and 16 (12.12%) of the respondents own a portable wireless device through which they access internet. It can also be noted from the study that a majority of the respondents, 105 (79.55%) uses a smartphone frequently in comparison with a laptop, desktop computer or tablet to access the internet. It can also be noted that 119 (90.15%) of the respondents accessed internet at least once a day and a very small percentage of 5.3, accessed internet once a week.

Perception of Teachers towards ICT integration

The third section of the questionnaire gives a clear indication that teachers are well aware of the usefulness of ICT to increases the quality of teaching and learning experience. Majority of the teachers with the mean score of 1.87, believes that technology enabled learningcreates more space for collaborative learning, which is beneficial for students and teachers alike.

The results also reveal that 72 (54.55%) of the respondents believe that technology-enabled learning can solve many of the problems prevailing in the current educational system. With a mean of 1.66, a very small percentage 9 (6.82%) are unable to predict whether technology implementation in teaching will lead to new opportunities for organising teaching and learning. On the other hand, a vast majority of 53 (40.15%) feel that technology will definitely open new horizons of opportunity for both teachers and students. Only 0.76% of the respondents agree that ICT cannot engage learners compared to other forms of learning. This also overwhelmingly proves the trust that majority of teachers has on the positive impact that ICT can yield if implemented efficiently. On the other hand, 6.06% of the respondents neither agree nor disagree to the notion that technology-based learning increases access to education and training.

Factors Motivating Teachers to use Technology-Enabled Tools for Teaching

The study also tried to understand the different factors that motivate teachers to adopt ICT tools for instruction in classrooms. With a mean score of 2.10, majority of the respondents feel that personal interest to use technology is a strong motivator to adopt and learn ICT tools. Self-gratification is a strong motivator for using ICT enabled tools with a mean score of 2.34. However, 41.67% of the teachers are not finding the training sessions that they receive as a very strong motivator to use them more frequently. 30.30% of the respondent's probe to the notion that better internet bandwidth at workplace can be a strong motivator to use ICT based tools for instruction with 2.40 mean score. The survey also shows that out of the total respondents, as less as 13.64% feel that credit towards promotion is a very strong motivator for engaging more with ICT tools. Peer recognition, prestige and status is also an average motivator for 52.27% of the respondents with a 2.52 mean score. It is also interesting to note that majority of the teachers who took part in the study

feels that reduction in existing workload could facilitate efforts leading to increased hours spend on learning and equipping themselves to use ICT tools for teaching.

Barriers to use Technology-Enabled Tools for Teaching

With a mean score of 2.69, majority of the respondents (59.85%) thinks that concerns about more workload, if they are an expert in ICT is only an average barrier ahead of them. Lack of training on technology-enabled learning is a strong barrier for 25% of the respondents to comprehensively implement ICT into their teaching pedagogies with a 2.66 mean score. Another significant barrier that teachers identified were the lack of technical assistance provided at their schools to which 41.67% of the respondents agreed that it is a strong barrier encountered with a 2.67 mean score. Lack of time to be more involved and upskill their existing knowledge in ICT is however found to be a average motivator for 53.79%, with a 2.66 mean score.

However, it was also noted that concerns over quality of E-content was a strong barrier in the consumption of online content for 34.85% with a 2.69 mean score. 21.97% of the teachers felt that lack of interest by students is a weak motivator and that they are not largely swayed by the disapproval of students. Mistrust on internet was observed to be a strong barrier for 37.88% of the respondents with a 2.64 mean score. Out of the total respondents 42.42% noted that inadequate IT infrastructure at their schools is a strong barrier for them to enhance their knowledge in Technology based teaching. Poor internet connectivity and networking within their institutions were also identified as a strong barrier for 37.88% of the respondents with 2.55 mean score.

Another interesting observation from the survey was that, 32.58% of the respondents were still facing psychological difficulties to shift from traditional way of teaching with a 2.85 mean score. 32.58% also noted that lack of institutional policy for technology-enabled learning is also a strong barrier for the adoption of technology enabled teaching.

Reliability Testing

Internal consistency of the instrument used for the study is tested using Cronbach's Alpha reliability testing. Since for there are three sets of Likert scales used for the study, each set is tested separately ranging from Strongly agree = 1, Agree = 2, Neither agree nor disagree = 3 and Disagree = 4 for the first set. For the second set, Very strong motivator = 1, Strong Motivator = 2, Average Motivator = 3, Weak motivator = 4 and Very weak motivator = 5. For set 3, Very Strong Barrier = 1, Strong Barrier = 2, Average Barrier = 3, Weak Barrier = 4 and Very Weak Barrier = 5. If the accepted value of Cronbach's Alpha is greater than 0.6 then the instrument is acceptable Kline (1999).

For the questions to identify the perception of teachers towards ICT integration, the value of Cronbach's alpha is .936, which is greater than 0.6. Hence the items in this set are acceptable and the instrument can be considered as valid. For the questions that identify the factors motivating teachers to use Technology-Enabled Tools for Teaching the value of Cronbach's alpha is .918 which means that the items in the instrument set is acceptable and satisfactory. Thus, it can be acceptable as a instrument for the research. For the section to understand the barriers to use technology-enabled tools for teaching, the reliability test shows that Cronbach's alpha value is .927, which is greater than 0.6 and thus the items on the instrument set are acceptable.

VI. DISCUSSION AND CONCLUSION

The results of the study clearly indicate that ICT enabled teaching pedagogies have more impact over traditional face-to-face lecture method. The teachers who were part of the study have also accepted the multi-dimensional approach of ICT to enable a mutually intersecting learning environment that is beneficial of both teachers and students alike. The participants have also identified that technology-enabled learning have the potential to saves time and effort for both teachers and students and thereby increase the efficiency.

Technology-enabled learning can open the gateways for collaborative learning. This would also increase the accessibility of students to different resource materials that are were earlier unavailable to them. The participants also strongly believe that ICT enabled problem solving methodologies can solve many of the existing challenges to the educational system. The potential of ICT to engage more learners as compare to other methodologies is also acknowledged by a majority of the participants. It is also important to note that the participants at large believe in the ability of ICT enabled teaching pedagogies to improve the quality of teaching at the teachers end and to improve the learning experience of the students.

In conclusion, if ICT integration is adequately planned and executed at all levels of its implementation, it has immense potential to transform the teaching and learning experience in primary schools. But it would also require all the stakeholders in the process to be efficient and adequately trained to carry on their assigned duties. Since, the landscape is under real-time upgradation teachers should also be exposed to such an environment where they are motivated to experiment and thereby learn more about the usage of tools on hand.

Finally, ICT integration is critical as a step to increase the competency of the educational system in the

country. There should also be advocacies at different levels aimed to improve the quality and also be keep a check at the existing paradigms from time to time.

LIMITATIONS OF THE STUDY

This study tries solely to understand the point of view of the teachers, who were also the key participants of the study. A different approach wherein student feedbacks are collected and their recommendations are taken into consideration would provide additional insights on how to improve the efficiency of ICT implementation. A subject specific analysis will further provide deeper understanding of the requirements for each subject. This study does not compartmentalise schools that are run by government and private bodies. However, such categorisation can also provide more insights on the way ICT implementation is carried out and how it could be improvised.

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